

Installing R and Setting up VSCode with Jupyter Notebook Guide

Introduction

This tutorial covers the installation of R, Jupyter Notebook, and VSCode configuration for R development across different platforms:

- Windows
- Linux
- WSL (Windows Subsystem for Linux)

Part 1: Installing R

1. Windows

Step 1: Download and Install R

- direct download from: <https://cran.r-project.org/bin/windows/base/>

Step 2: Download and Install RStudio (optional)

- Download from: <https://posit.co/download/rstudio-desktop/>

Step 3: Verify Installation

```
# In Command Prompt
R --version

# In PowerShell
R.exe --version
```

LANGUAGE-POWERSHELL

Troubleshooting: If you get "R is not recognized" error:

This means R is not added to your system PATH. Here are the solutions:

Solution : Add R to PATH (Recommended)

0. Find R Installation Path

```
# Find where R.exe is Located (standard path)
Get-ChildItem -Path "C:\Program Files\R" -Recurse -Name "R.exe"

# If R is installed elsewhere, search all drives (may take time)

# In PowerShell, use:
where.exe /R C:\ R.exe

# In Command Prompt, use:
where /R C:\ R.exe
```

LANGUAGE-POWERSHELL

💡 **Note:** If you installed R in a custom directory (not the default `C:\Program Files\R`), replace the path in the above commands with your actual installation directory.

In PowerShell, use `where.exe` (not just `where`) to avoid conflicts with PowerShell's `Where-Object` cmdlet.

1. Press **Windows Key + R** → type `sysdm.cpl`
2. Go to **Advanced** tab → **Environment Variables**
3. In **System Variables**, select **Path** → **Edit**
4. Click **New** and add: `$R_PATHS`
5. Close and reopen PowerShell, then test: `R --version`

💡 **Note:** Ensure you are replacing `$PATHS` with `$R_PATHS` and it's not included `/R.exe`.

2. Linux (Ubuntu/Debian)

Step 1: Update System

```
sudo apt update
sudo apt upgrade
```

LANGUAGE-BASH

Step 2: Install R

```
# Ubuntu/Debian
sudo apt install r-base r-base-dev
```

LANGUAGE-BASH

Step 3: Install Additional Packages

```
sudo apt install build-essential libcurl4-openssl-dev libssl-dev libxml2-dev
```

LANGUAGE-BASH

3. WSL (Windows Subsystem for Linux)

Step 1: Enable WSL

```
# In PowerShell with Administrator privileges
wsl --install
```

LANGUAGE-POWERSHELL

Step 2: Install Ubuntu in WSL

```
wsl --install -d Ubuntu
```

LANGUAGE-POWERSHELL

Step 3: Install R in WSL

```
# Enter WSL environment and run Linux commands
sudo apt update
sudo apt install r-base r-base-dev build-essential
```

LANGUAGE-BASH

Part 2: Installing Python and Jupyter (All Platforms)

Step 1: Install Python

```
# Windows - from Microsoft Store or python.org

# Linux
sudo apt install python3 python3-pip

# WSL - same as Linux
sudo apt install python3 python3-pip
```

LANGUAGE-BASH

Step 2: Install Jupyter

```
pip3 install jupyter jupyterlab notebook
```

LANGUAGE-BASH

Step 3: Install IRKernel (R Kernel for Jupyter)

First, open R

```
./R.exe
```

LANGUAGE-BASH

now In R environment, run these commands:

```
install.packages('IRkernel')

# for system-wide installation
# Installs the R kernel for all users on the system
IRkernel::installspec(user = FALSE)

# or

# for user installation
# Installs the R kernel only for the current user
# Kernel files stored in user's home directory
IRkernel::installspec()

# Check is installation successful
# it's successful if you don't receive any warning or error
library(IRkernel)

# Exit R
quit()
```

LANGUAGE-R

💡 **Note:** If prompted to select a CRAN mirror, choose option 1 (Global/Cloud) or select a mirror close to your location for faster downloads.

💡 **Note:** If asked about personal library (Windows), type `yes`. This happens when you don't have admin rights to install in system directory.

Verify IRkernel Installation:

check if R kernel is registered with Jupyter

```
jupyter kernelspec list
```

LANGUAGE-BASH

You should see output similar to:

```
Available kernels:
  ir          C:\Users\YourName\AppData\Roaming\jupyter\kernels\ir
  python3     C:\Users\YourName\AppData\Local\Programs\Python\Python311\share\jupyter\kernels\python3
```

If you see the `ir` kernel listed, IRkernel installation was successful!

Part 3: Setting up VSCode

Step 1: Install VSCode

- Download from: <https://code.visualstudio.com/>

Step 2: Install Essential Extensions

Install these extensions in VSCode:

1. Python (Microsoft)
2. R (REditorSupport)
3. Jupyter (Microsoft)
4. Jupyter Keymap (Microsoft)
5. Jupyter Notebook Renderers (Microsoft)
6. R Tools (optional)

Step 3: Configure R Path

Windows:

In settings.json add this:

```
{  
  // R Path (adjust version number to match your installation)  
  "r.rpath.windows": "C:\\Program Files\\R\\R-4.5.1\\bin\\x64\\R.exe",  
  "r.terminal.windows": "C:\\Program Files\\R\\R-4.5.1\\bin\\x64\\R.exe",  
  
  // Language Server (for IntelliSense)  
  "r.lsp.enabled": true,  
  "r.lsp.lang": "en",  
  
  // IntelliSense  
  "r.completion.enabled": true,  
  "r.completion.showFunctionDefinition": true,  
  
  // Editor  
  "editor.suggest.showSnippets": true,  
  "editor.tabCompletion": "on", // (optional)  
}
```

LANGUAGE - JSON

Linux/WSL:

```
// In settings.json  
{  
  "r.rpath.linux": "/usr/bin/R",  
  "r.terminal.linux": "/usr/bin/R"  
}
```

LANGUAGE - JSON

Test IntelliSense

Create a test `.R` file and try bellow code. (IntelliSense should show suggestions)

```
mtcars$ # Should show column names  
data.  # Should show data-related functions
```

LANGUAGE - R

Part 4: Testing and Usage

Step 1: Create New Jupyter Notebook

1. In VSCode, press `Ctrl + Shift + P`
2. Select `Jupyter: Create New Jupyter Notebook`
3. In the top right corner, select R kernel

Step 2: Test R Code

```
# Cell 1 - Basic test
print("Hello, World!")
R.version.string

# Cell 2 - Package test
install.packages("ggplot2", quiet = TRUE)
library(ggplot2)

# Cell 3 - Simple plot
data(mtcars)
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Weight vs Fuel Economy",
       x = "Car Weight",
       y = "Miles per Gallon")
```

LANGUAGE-R

Step 3: Save and Run

- `Ctrl+S` : Save notebook
- `Shift + Enter` : Run current cell
- `Ctrl + Enter` : Run cell without moving to next

Tips and Recommendations

Additional VSCode Settings

```
{
  "jupyter.askForKernelRestart": false,
  "jupyter.interactiveWindow.creationMode": "perFile",
  "r.bracketedPaste": true,
  "r.sessionWatcher": true
}
```

LANGUAGE-JSON

Recommended Packages

```
# Essential packages
install.packages(c(
  "tidyverse",    # Data analysis toolkit
  "data.table",   # Big data handling
  "plotly",       # Interactive plots
  "shiny",        # Web applications
  "knitr",        # Report generation
  "rmarkdown"    # R Markdown
))
```

LANGUAGE-R

Additional Resources

- [R Project Official](#)
- [Jupyter Documentation](#)
- [VSCode R Extension](#)
- [IRKernel GitHub](#)

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💡 **Note:** After completing the installation, restart VSCode to apply all changes.